

ค่าเฉลี่ย (Mean) $\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$

Sample Variance $\frac{1}{n-1} (\sum_{i=1}^n x_i^2 - n(\bar{x})^2)$

Ex) 45 49 55 41 62

$$\sum x_i^2 = (45^2 + 49^2 + 55^2 + 41^2 + 62^2)$$

$$= \frac{1}{4} (12976 - (5 \times (50.4)^2))$$

$$= 68.8 \#$$

ค่าเบี่ยงเบน (STD) $s = \sqrt{\text{Variance}}$

Ex) $\sqrt{68.8} = 8.2946 \#$

Modifi (x + s.mean by c) $\bar{Y} = a\bar{X} + b$

• Variance $s_y^2 = a^2 s_x^2 \cdot 940 \quad s_y = 1a1s_x$

มัธยฐาน (Median) Odd = $(n+1)/2$

* Cori * Even = $\text{ค่า } \bar{x} \text{ ของ } n/2 \text{ กับ } (n/2)+1$

Ex) 41 45 49 55 62 705

∴ $((6/2) + ((6/2) + 1)) / 2$

$= (49 + 55) / 2 = 52 \#$ * นาคำดับ

* Quartile (if "Int" else left/right)

$Q_{1,2,3} = 0.25(n+1), 0.50, 0.75$ * Mean?

Percentile (if "Int" else left/right)

Ex) $n = 15$ and $p = 20$

∴ $0.20(15+1) = 3.2$ (use data 3, 4)

$(1640 + 2313) / 2 = 1976.5 \#$ * Mean

ความหนาแน่น (Density) * Histogram *

Relative Freq / Class Width

Ex) temp $56^\circ - 60^\circ$, RF = 0.05

∴ Density = $0.05 / 5 = 0.01 \#$

Boxplot (Quertile) กรงนก / กรงจาย/outl.

Normal Distribution (Z-TABLE)

* mean (μ) VAR (σ^2) std (σ)

$z = (x - \mu) / \sigma$

Ex) $\mu = 5$, $\sigma^2 = 9$

a) $P(X \leq 1) = \Phi((1-5)/3) = \Phi(-1.33)$

$= 1 - 0.9082 = 0.0918 \#$

b) $P(X > 5.5) = 1 - \Phi((5.5-5)/3) = \Phi(0.17)$

$= 1 - 0.5675 = 0.4325 \#$

c) $P(2 \leq X \leq 6) = \Phi(0.33) - (1 - \Phi(1))$

$= 0.6293 - (1 - 0.8413) = 0.4706 \#$

CLT (แจกแจงให้ n หนา) $(x - \bar{x}) / (s/\sqrt{n})$

Ex) $\bar{x} = 22$ $s = 10$ $n = 60$, $P(X < 23)$?

$P(x < (23 - 22) / (10/\sqrt{60}))$

$= \Phi(0.63) = 0.7357 \#$

ช่วงความเชื่อมั่น (C.I.) $\bar{X} \pm z_{\alpha/2} \cdot (s/\sqrt{n})$

Ex) $n = 80$ $\bar{x} = 250$ $s = 15$, 95%

$a = 1 - 0.95 = 0.05$ $a/2 = 0.025$

$z_{0.025} = 1 - 0.025 = 1.96$

$250 \pm 1.96 \cdot (15/\sqrt{80}) = (-, +)$

Ex) $n = 66$ $\bar{x} = 116$ $s = 12$, (108.26, 121.34)

① $121.34 = 116 + x \cdot (12/\sqrt{66})$ * ข้างขว

② $x \approx 3.385$ $z_{\alpha/2} = 3.39$ $a/2 = 0.9997$

③ $\text{หา } a \rightarrow 1 - \alpha/2 = 0.9997 = 0.9994$ * ข้างขว

Hypothesis $z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

Ex) $n = 4$ $\bar{x} = 50$ $s = 4$ $\mu = 51$

t-table df = $n - 1 = 3$

$t = (50 - 51) / (4/\sqrt{4}) = -1/2 = -0.5$

P-Value = $0.25 < P\text{-Value} < 0.40$

Regression ($r = 0 \rightarrow$ Uncorrelated)

$r = (1/(n-1)) \sum_{i=1}^n ((x_i - \bar{x}) / s_x) ((y_i - \bar{y}) / s_y)$

$s_{x,y} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)}}$

ค่าการถดถอย

$y = \hat{\beta}_0 + (\hat{\beta}_1)x$

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$\hat{\beta}_1 = \frac{\sum_{i=1}^n ((x_i - \bar{x})(y_i - \bar{y}))}{\sum_{i=1}^n (x_i - \bar{x})^2}$

Example zone 1

PROBABILITY [$\bar{x} = 22$ $\sigma = 10$]

1) $20 < X < 23$

$\text{Sol}^N \quad x = (20 - 22) / 10 = 1 - \Phi(0.2)$

$= 1 - (1 - 0.5793) = 0.5793$

$x = (23 - 22) / 10 = \Phi(0.1)$

$= 0.5398$

∴ $0.5398 - 0.5793 = 0.0395 \#$

2) $X > 19$

$\text{Sol}^N \quad x = (19 - 22) / 10 = 1 - \Phi(0.3)$

$= 1 - (1 - 0.6179) = 0.6179 \#$

3) Percentile 60 ($\rightarrow 0.6000$)

$\text{Sol}^N \quad \bar{x} + z\text{-score} \cdot \sigma = 22 + 0.25 \cdot 10$

$= 24.5 \#$

4) $X < 23$ $n = 40$

$\text{Sol}^N \quad (23 - 22) / (10/\sqrt{40}) = \Phi(0.63)$

$= 0.7357 \#$

C. Interval [$n = 90$ $\bar{x} = 252$ $\sigma = 17$]

$SE = \sigma / \sqrt{n} \rightarrow 17 / \sqrt{90} = 1.792$

1) (249.3701, 254.6502)

$\text{Sol}^N \quad 252 - 249.3701 = 2.6299$

$\rightarrow 2.6299 / 1.792 (SE) = \Phi(1.47)$

$= 0.9292 \rightarrow 92.92 \%$

$\rightarrow 100 - 92.92 = 7.08$

$\rightarrow 92.92 - 7.08 = 85.84 \%$ #

2) 85%

$\text{Sol}^N \quad 100 - 85 = (15/2) + 85 = 92.5$ * z-table

∴ $252 \pm (1.44 \cdot 1.792 (SE))$

$= (249.4195, 254.5805) \#$

Hypothesis [$\bar{x} = 98$ $s = 6$ $n = 6$]

$H_0: \mu \geq 99$ $H_1: \mu < 99$

1) test statistic

$\text{Sol}^N \quad z = (98 - 99) / (6/\sqrt{6}) = -0.4082$

∴ P-value = $1 - 0.6591 = 0.3409 \#$

2) 7% (0.07) = ไม่สามารถปฏิเสธ

3) 3% (0.03) = "

* ถ้า P-value > ระดับนัยสำคัญ = ไม่ปฏิเสธ

Note:

z-score neg (-)

$\hookrightarrow P(T < 50)$

Ex) 25 (0.2500)

$\rightarrow 0.7500$

* z-table

$\bar{x} + (-z \cdot s)$

C.I. Level (2 ข้าง)

68% = 1

90% = 1.645

95% = 1.96

99% = 2.58

99.7% = 3

ข้างเดียว ↓

90% = 1.28

95% = 1.645

99% = 2.33

* ข้อยกเว้น < 30 $\rightarrow$ t-table

$\bar{x} \pm t_{n-1, \alpha/2} \cdot (s/\sqrt{n})$

Setup

ON $\rightarrow$ SHIFT $\rightarrow$ 9 (Reset) $\rightarrow$ 1 (Setup) $\rightarrow$ = $\rightarrow$ AC

Cal

Menu $\rightarrow$ 2 (Graph) $\rightarrow$ $y = a + bx$

$\rightarrow$ 6 (b1) (b2) $\rightarrow$ = $\rightarrow$ AC

SHIFT OPTN

$\hookrightarrow$ 7 OPTN $\rightarrow$ 3 (Regress)

SHIFT

$\hookrightarrow$ 7 SHIFT $\rightarrow$ 1 $\rightarrow$ Regress